



Polygons

A CLOSED TWO-DIMENSIONAL SHAPE OF THREE OR MORE SIDES.

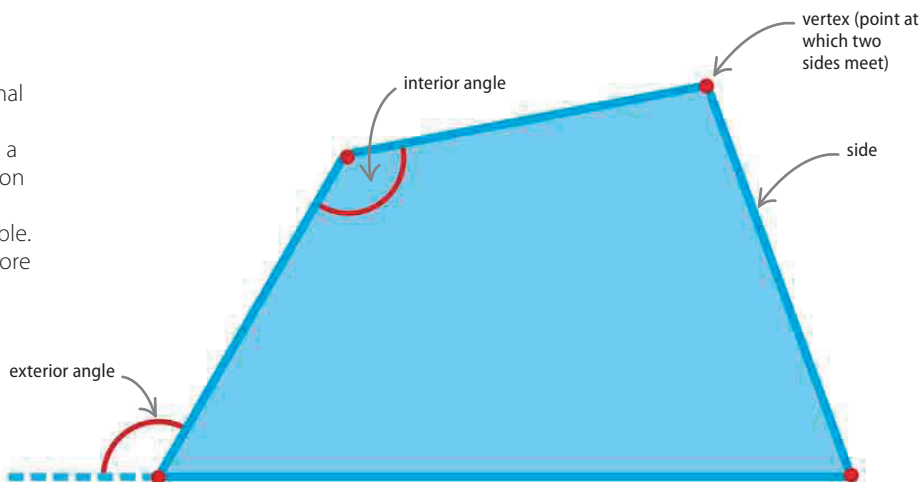
Polygons range from simple three-sided triangles and four-sided squares to more complicated shapes such as trapezoids and dodecagons. Polygons are named according to the number of sides and angles they have.

What is a polygon?

A polygon is a closed two-dimensional shape formed by straight lines that connect end to end at a point called a vertex. The interior angles of a polygon are usually smaller than the exterior angles, although the reverse is possible. Polygons with an interior angle of more than 180° are called concave.

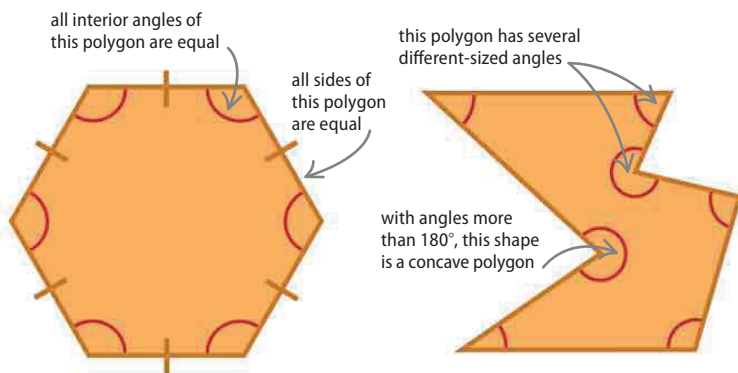
Parts of a polygon

Regardless of shape, all polygons are made up of the same parts—sides, vertices (connecting points), and interior and exterior angles.



Describing polygons

There are several ways to describe polygons. One is by the regularity or irregularity of their sides and angles. A polygon is regular when all of its sides and angles are equal. An irregular polygon has at least two sides or two angles that are different.



Regular

All the sides and all the angles of regular polygons are equal. This hexagon has six equal sides and six equal angles, making it regular.

Irregular

In an irregular polygon, all the sides and angles are not the same. This heptagon has many different-sized angles, making it irregular.

SEE ALSO

◀ 84–85 Angles

◀ 116–117 Triangles

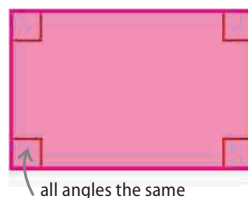
◀ 120–121 Congruent triangles

◀ 130–133 Quadrilaterals

LOOKING CLOSER

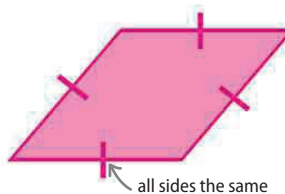
Equal angles or equal sides?

All the angles and all the sides of a regular polygon are equal—in other words, the polygon is both equiangular and equilateral. In certain polygons, only the angles (equiangular) or only the sides (equilateral) are equal.



◁ Equiangular

A rectangle is an equiangular quadrilateral. Its angles are all equal, but not all its sides are equal.



◁ Equilateral

A rhombus is an equilateral quadrilateral. All its sides are equal, but all its angles are not.